WIL Project — Milestone 1

Test-Driven RAG for Australian Individual Tax Deductions

1. Group ID

Project RAG

2. Trello Board

Trello Board:
https://trello.com/invite/b/6a851b4659b5cda68b58b730/ATTIea1b1bbb0d8b083495781309745f6c20B0AC80AE/project-rag

3. GitHub Repository

GitHub Repo: https://github.com/ShreeshJoshi7/rag-tax-deductions

4. Project Aim Statement

We're building a RAG-based assistant that helps Australian individual taxpayers work out which income tax deductions they're eligible for, with every answer grounded in official ATO guidance. We'll judge success mainly on how faithful the answers are to the source material and how often the system has to admit it doesn't know.

User Stories

- As an individual taxpayer, I want to ask what work-related expenses I can claim so I don't miss out on deductions I'm entitled to.
- As a first-time filer, I want ATO terminology explained in plain English so I can understand my obligations without having to pay a tax agent.
- As a taxpayer, I want every answer to cite its ATO source so I can actually check where the information came from.

5. Group Members

Student ID	Full Name	Role / Skills / Strengths	% Contribution*
s4206111	Shreesh Anil Joshi	RAG Pipeline— retrieval, embeddings, generation, vector DB integration. Renewable-energy ML/anomaly-detection background.	20%
s4149947	Athith M S	Evaluation Engineer — builds evaluation scripts (NDCG, faithfulness, unanswered-question rate).	20%
s4164152	Sahana Nataraja	Data Engineer — scripts to source, clean, and chunk ATO documents into the knowledge base.	20%
s4234052	Brunda Shankar	Frontend Engineer — builds and codes the Streamlit interface, wires it to the RAG pipeline.	20%
s4159521	Vaishnavi Sreenath	Integration & Documentation — repo structure, branch merges, keeps requirements.txt and setup docs reproducible for the team, and builds a script to summarise evaluation results, also coordinates Trello and milestone reporting.	20%

**Contribution is shown as equal at this stage because Milestone 1 work so far has been mostly collaborative planning (aim, roles, stack, repo/Trello setup) rather than individual technical output. One member led the initial pipeline reproduction on the team's behalf so the group has a working reference to build from in Week 1; from Week 2 onward, contribution will be tracked and reported per person based on actual technical output.*

6. Progress So Far

We settled on Australian individual income tax deductions as our project domain and are using publicly available ATO guidance documents as the knowledge base. We kept this deliberately small, the brief encourages a tight, high-quality document set over a huge scraped collection, and the documents felt like a corpus we could actually work with properly in the time we have.

As a group, we met to nail down the project aim, user stories, and role split (see Section 5), set up the shared GitHub repo and Trello board, and agree on our tech stack. After that, we took on the first piece of hands-on technical work for the team: reproducing the RAG pipeline and evaluation framework from Walert (RMIT's FAQ Open Day chatbot, github.com/rmit-ir/walert) end-to-end on its original dataset. The goal was to have one working,

validated pipeline the whole team could build from in Week 1, rather than everyone hitting the same environment setup issues on their own. That work involved:

- Setting up a Python 3.9 environment through WSL and working through the dependency stack — PyTorch, Transformers, FAISS, Pyserini, spaCy, pytrec_eval.
- Encoding the FAQ knowledge base with a dense passage encoder (TCT-ColBERT), then building both a BM25 (Lucene) index and a dense FAISS index.
- Running retrieval over all the evaluation topics and scoring it against Walert's ground-truth relevance judgments using NDCG.

Here's how Walert's original intent-based system compares to RAG-based retrieval on NDCG@1/3/5:

Model	NDCG@1	NDCG@3	NDCG@5
Walert (intent-based)	0.083	0.039	0.039
RAG (BM25)	0.167	0.257	0.329
RAG (Dense / FAISS)	0.250	0.238	0.238

Both RAG-based approaches clearly beat Walert's original intent-based system across every NDCG cut-off. That's a good sign that it confirms RAG is the stronger retrieval strategy for this kind of task and gives us confidence in the reference pipeline before we adapt it to the tax deductions knowledge base.

7. Plan for the Next Three Weeks

- Week 1: Source and clean the ATO deduction guidance documents, build the tax-domain versions of topics.csv, collection.csv, groundtruth.csv, and qrels.txt, and adapt the encoding/indexing pipeline to the new corpus.
- Week 2: Hook up an LLM for answer generation over the retrieved passages (OpenAI, Claude API, or a free local model via Ollama, which still to be confirmed with our mentor), and put together an initial evaluation question set with ground-truth answers.
- Week 3: Build out the evaluation framework — NDCG, % of unanswered questions, and at least one more dimension such as faithfulness and put together an initial Streamlit demo.

8. Bottlenecks and Risks

- Chunking tax documents that are full of tables and numeric thresholds, without losing accuracy in the process.
- Putting together a fair, large-enough evaluation question set when there's no existing labelled dataset for the tax domain to lean on.
- Getting confirmation from the mentor on whether we should use a paid LLM API or a free local model (Ollama), this affects the whole team's stack, so we need it settled early.
- Keeping parallel work coordinated across five team members with different roles and different levels of experience.

References

Spina, D., Trippas, J. R., Cavedon, L., & Sanderson, M. (2024a). *Walert: Putting conversational information seeking knowledge into action by building and deploying a large language model-powered chatbot*. Proceedings of the 2024 ACM SIGIR Conference on Human Information Interaction and Retrieval. https://doi.org/10.1145/3627508.3638309

Australian Taxation Office. (n.d.). *Deductions you can claim*. https://www.ato.gov.au/individuals-and-families/income-deductions-offsets-and-records/deductions-you-can-claim

Spina, D., Trippas, J. R., Cavedon, L., & Sanderson, M. (2024b). *Walert* [Source code]. GitHub. https://github.com/rmit-ir/walert